Department of Electrical and Computer Engineering
North South University

Directed Research

PROACTIVE BITRATE ADAPTATION FOR RESILIENT CLOUD GAMING OVER UNSTABLE 5G NETWORKS

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Spring, 2026

LETTER OF TRANSMITTAL

October, 2026

To
Chairman,
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: Submission of Directed Research Report on "Proactive Bitrate Adaptation for Resilient Cloud Gaming over Unstable 5G Networks"

Dear Sir,

With due respect, we would like to submit our Directed Research Report on "Proactive Bitrate Adaptation for Resilient Cloud Gaming over Unstable 5G Networks" as a part of our BSc program. The report presents the design and evaluation of a proactive bitrate adaptation framework for cloud gaming over unstable 5G networks, using real-world 5G measurements and Long Short-Term Memory based throughput forecasting.

This research was valuable to us because it provided practical experience in applying machine learning, networking concepts, and Quality of Experience analysis to a realistic communication problem. We tried to meet the academic and technical requirements of the directed research report as carefully as possible.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative.

Sincerely Yours,

Tamanna Akter Mou
ECE Department
North South University, Bangladesh
.....
Sanin Tabassum
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North South University, Bangladesh

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Tashfiq Mahmud Niloy
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APPROVAL

The undersigned students from the Department of Electrical and Computer Engineering of North South University have worked on the Directed Research Project titled "Proactive Bitrate Adaptation for Resilient Cloud Gaming over Unstable 5G Networks" under the supervision of the faculty advisor for partial fulfillment of the requirement for the degree of Bachelor of Science in Engineering, and the work has been submitted for evaluation.

Supervisor's Signature

.....

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Chairman's Signature

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DECLARATION

This is to declare that this project/directed research is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor.

Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' Names & Signatures

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2. Sanin Tabassum _____

3. Tashfiq Mahmud Niloy _____

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The authors also thank the Department of Electrical and Computer Engineering, North South University, Bangladesh, for facilitating the research. We are grateful to our classmates, friends, and family members for their support and encouragement throughout the project.

ABSTRACT

Proactive Bitrate Adaptation for Resilient Cloud Gaming over Unstable 5G Networks

Cloud gaming over fifth-generation (5G) millimeter-wave networks faces a fundamental challenge - the extreme volatility of mmWave signals causes sudden throughput drops that reactive Adaptive Bitrate algorithms cannot address fast enough to prevent Motion-to-Photon latency spikes and gameplay disruption. This paper presents a proactive bitrate adaptation system for resilient cloud gaming over unstable 5G networks, built around a Long Short-Term Memory neural network trained to forecast 5G downlink throughput before degradation reaches the client. The LUMOS 5G dataset, containing 68,118 real-world mmWave measurements collected across walking and driving mobility scenarios on a commercial Verizon 5G network, was used to train and evaluate the system. A two-layer LSTM model with 64 hidden units was trained for 50 epochs using seven UE-side telemetry features including signal strength indicators, device speed, and historical throughput. The model achieved a Mean Squared Error of 0.007447 and Mean Absolute Error of 0.053223 on the test set, outperforming a Simple Moving Average baseline by 61.2% on MSE. A threshold-based bitrate adaptation policy applied to the predictions maintained gaming quality at 720p or above in 88.9% of test cases while reducing unnecessary bitrate switching frequency from 27.87% to 19.34% compared to reactive adaptation. Handover analysis identified 1,371 cell tower transition events with an average throughput drop of 85.23 Mbps, confirming the necessity of predictive adaptation in urban cloud gaming deployments. The results demonstrate that LSTM-based proactive throughput prediction is a viable and effective foundation for resilient cloud gaming QoE optimization over volatile 5G networks.

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Chapter 1 Introduction

1.1 Background and Motivation

The evolution of telecommunication technologies, particularly the rollout of fifth-generation (5G) networks, has revolutionized the landscape of mobile computing and digital entertainment. Among the most demanding and rapidly growing applications enabled by 5G is Cloud Gaming. Unlike traditional gaming, where rendering and processing are handled by local hardware, cloud gaming offloads these computationally intensive tasks to remote servers in data centers or edge nodes. The rendered video frames are then streamed over the internet to thin clients, such as smartphones, smart TVs, or low-power personal computers, while the player's input commands are sent back to the server in real-time [5]. This paradigm shift democratizes access to high-fidelity gaming experiences by removing the barrier of expensive hardware. However, it imposes stringent requirements on the underlying network infrastructure, specifically demanding ultra-low latency and consistently high bandwidth to ensure a seamless Quality of Experience (QoE).

Despite the theoretical capabilities of 5G networks, including millimeter-wave (mmWave) spectrum utilization that offers multi-gigabit throughput, practical deployments face significant challenges. 5G mmWave signals are highly susceptible to physical obstructions, atmospheric absorption, and user mobility. A user walking behind a building or experiencing a cell tower handover can induce sudden and severe throughput fluctuations and signal fading [2]. For standard video streaming platforms like Netflix or YouTube, these network anomalies are effectively masked by maintaining large playback buffers. Cloud gaming, however, fundamentally operates under a strict "Gaming Constraint" - because the system is highly interactive, the luxury of large buffers is non-existent. Any Motion-to-Photon (MTP) latency exceeding a few tens of milliseconds drastically degrades the user's gameplay experience [1].

To cope with varying network conditions, multimedia streaming relies on Adaptive Bitrate (ABR) algorithms [9]. Traditional ABR algorithms operate reactively - they monitor historical network throughput and adjust the video encoding bitrate for subsequent chunks accordingly. While effective for buffered streaming, these reactive approaches are fundamentally mismatched with cloud gaming over 5G. By the time a reactive ABR algorithm detects a sudden throughput drop, the congestion has already occurred and the user experiences a noticeable stutter or game freeze [9]. Recent research has pivoted towards Deep Reinforcement Learning (DRL), utilizing architectures such as the Asynchronous Advantage Actor-Critic (A3C), which have demonstrated superior capabilities in learning complex relationships between network states and optimal bitrate decisions

[3]. However, many existing DRL-based ABR models remain inherently reactive, failing to account for the catastrophic impact of sudden MTP latency spikes during high-motion gameplay [12]. Fig. 1 illustrates the fundamental contrast between reactive and proactive ABR adaptation approaches in the context of cloud gaming over 5G networks.

Therefore, there is a critical need for a paradigm shift from reactive adaptation to proactive optimization. If a cloud gaming system can predict an impending 5G signal fade or a handover event just hundreds of milliseconds in advance, it can proactively downscale the video bitrate before the network capacity drops [7]. This pre-emptive action would prevent network queues from overflowing, thereby eliminating congestion-induced delay and preserving the ultra-low MTP latency essential for an acceptable gaming experience.

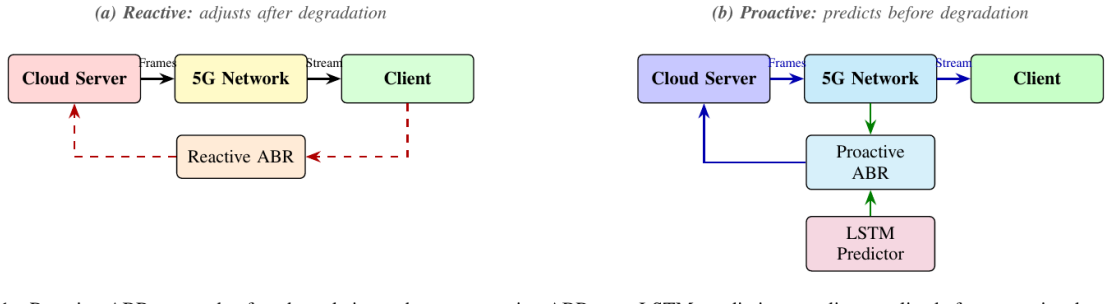


Fig. 1. Reactive ABR responds after degradation, whereas proactive ABR uses LSTM prediction to adjust quality before capacity drops [7], [10].

1.2 Purpose and Goal of the Project

The primary purpose of this research project is to design, implement, and rigorously evaluate a novel, proactive Adaptive Bitrate (ABR) streaming framework specifically tailored for resilient cloud gaming over unstable 5G networks. We aim to overcome the inherent limitations of conventional reactive ABR algorithms by introducing predictive intelligence directly into the bitrate adaptation control loop.

The central goal is to develop a specialized Deep Reinforcement Learning agent based on the A3C architecture, augmented with Long Short-Term Memory (LSTM) layers. This proposed model seeks to anticipate network degradation and adjust the video stream pre-emptively, safeguarding the stringent latency constraints of interactive cloud gaming. The project outlines three specific objectives:

1) Predictive Modeling for Network Telemetry: To develop an AI-based prediction system that forecasts 5G signal fading and throughput drops 100-300 milliseconds in advance, utilizing the LUMOS 5G Dataset containing real-world throughput and signal strength data across various mobility patterns [10].

2) QoE Optimization Prioritizing MTP Latency: To formulate a specialized QoE reward function that heavily penalizes MTP latency spikes and frame-dropping events over visual resolution, ensuring the DRL agent prioritizes responsiveness during high-motion gameplay [3].

3) Seamless Handover Management: To minimize stalling and video freezing during 5G cell tower transitions by proactively downscaling bitrate prior to handover execution and aggressively upscaling once the new connection stabilizes [2].

By achieving these objectives, this project contributes a significant advancement to multimedia streaming and mobile edge computing, demonstrating that predictive DRL can effectively mitigate the volatility of 5G networks and enable truly resilient, ultra-low latency cloud gaming experiences.

1.3 Organization of the Report

The remainder of the report is organized as follows. Chapter 2 reviews related work on ABR streaming, cloud gaming, and learning-based network adaptation. Chapter 3 describes the dataset, preprocessing pipeline, LSTM predictor, and bitrate-selection methodology. Chapter 4 presents the experimental results, baseline comparisons, mobility analysis, bitrate outcomes, and handover assessment. Chapter 5 summarizes the findings, discusses limitations, and outlines future work.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

5G networks and cloud gaming require multimedia streaming protocols that prioritize responsiveness. Large playback buffers can absorb network jitter in conventional video delivery but conflict with interactive gaming's latency constraints [5]. Network-aware application-layer adaptation therefore seeks to balance video quality, interaction latency, and playback continuity to improve Quality of Experience (QoE) [5].

Traditional Adaptive Bitrate (ABR) algorithms use rate-based, buffer-based, or hybrid heuristics [9]. Rate-based methods estimate throughput from previous chunk downloads, but historical measurements can become unreliable in volatile 5G millimeter-wave (mmWave) environments. Buffer-based methods such as BOLA select bitrates from playback-buffer occupancy, whereas cloud gaming's Motion-to-Photon (MTP) latency constraint of 50-150 ms limits buffering [9]. Hybrid methods combine throughput estimates with buffer awareness but remain dependent on prediction accuracy [9].

Deep Reinforcement Learning (DRL) formulates ABR control as a Markov Decision Process. Early work used Asynchronous Advantage Actor-Critic (A3C) to learn policies from experience [1]. To improve stability and generalization, SABR combines Behavior Cloning (BC) pretraining via Direct Preference Optimization (DPO) with Proximal Policy Optimization (PPO) fine-tuning. Despite improving average QoE, it selects subsequent chunk bitrates from past network states rather than explicitly forecasting impending 5G capacity drops [1].

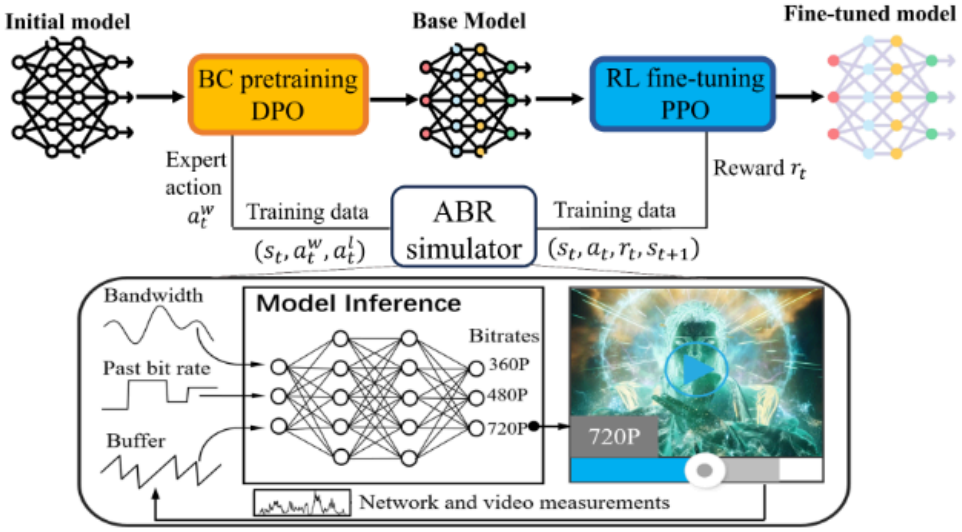


Fig. 2. The two-stage SABR framework utilizing Behavior Cloning pretraining and Reinforcement Learning fine-tuning. Adapted from [1].

5G New Radio (NR) mmWave links are sensitive to path loss and physical blockage. The "Reminis" architecture addresses transport-layer congestion control under the larger Bandwidth-Delay Product of 5G [2]. Its proactive slowdown module uses first-order linear regression on the delay derivative to reduce the congestion window before congestion occurs. Application-layer adaptation needs similar predictive intelligence to regulate video-frame sizes generated by the game engine [2].

Extended Reality (XR) imposes similar constraints. A Multi-Agent Reinforcement Learning (MARL) approach using Optimistic Mixture of Q-Values (oQMIX) and attention-weighted cell congestion metrics reduces packet loss and jitter across heterogeneous 5G streams [3]. However, decisions respond to current observations without forecasting future states. FairStream evaluations further show that standard PPO agents can pursue bandwidth aggressively, causing unfair allocation and rebuffering for competing users [4].

EnDASH-5G incorporates User Equipment energy efficiency into its DRL reward. An LSTM predicts download throughput from recent measurements and radio signal quality, feeding a cascaded A3C architecture that schedules chunk fetching to increase time in low-power Radio Resource Control (RRC) states [5]. This demonstrates predictive throughput modeling, but the objective is energy-aware scheduling rather than preventing MTP latency violations through proactive bitrate adaptation [5].

Application-layer ABR can lack visibility into transport conditions. The "eBandit" framework uses the extended Berkeley Packet Filter (eBPF) subsystem to monitor transport telemetry within the Linux kernel, enabling sub-millisecond awareness [6]. A lightweight Multi-Armed Bandit selects among ABR heuristics. Although this improves observability, it does not provide deep temporal prediction [6].

For 360-degree video, MADRL combines rate adaptation with a multimodal spatial-temporal attention Transformer to predict user viewports. Probabilistic trajectory maps address erratic head movement and reduce wasted bitrate [7]. Such forecasting principles are relevant, but cloud gaming instead requires predictions of wireless-channel changes [7].

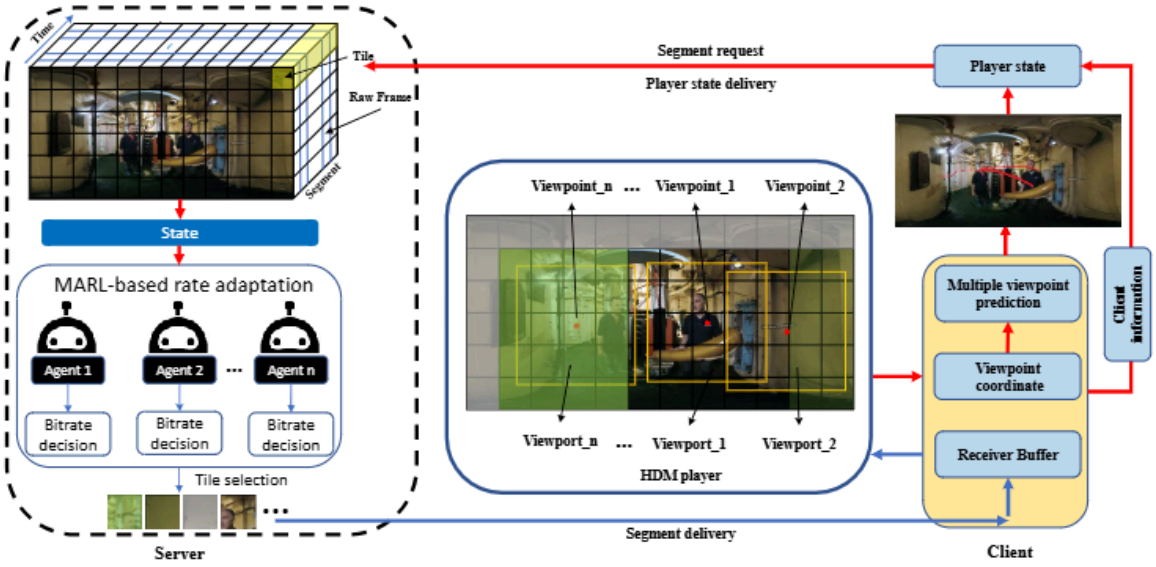


Fig. 3. System architecture of MADRL-based streaming integrating a spatial-temporal attention Transformer. Adapted from [7].

Mobile Edge Computing (MEC) reduces propagation delay by moving rendering closer to 5G users, making the variable last-mile wireless link a key bottleneck [8]. Within data centers, proactive Service Function Chain (SFC) provisioning combines Spatio-Temporal Graph Neural Networks and LSTMs to forecast resource utilization, reporting a 34.8% reduction in end-to-end cloud gaming latency. This supports forecast-driven control, although its scope is infrastructure provisioning rather than access-network bitrate adaptation [10].

Quantum Machine Learning (QML) has also been explored for QoE optimization. A hybrid of Variational Quantum Classifiers and Simulated Bifurcation improves accuracy and optimization speed over classical models [11]. However, static network snapshots do not supply the continuous temporal forecasts needed during gaming sessions [11].

SafeSABR addresses tail risk using a Conditional Value-at-Risk (CVaR) penalty during RL fine-tuning to discourage aggressive bitrates under throughput uncertainty [12]. A deterministic runtime auditor checks requested bitrates against a historical safe-capacity lower bound. This mitigates severe rebuffering, but backward-looking statistics cannot anticipate a capacity collapse before its precursors become observable [12].

The reviewed work motivates three research priorities: application-layer ABR that anticipates 5G degradation; QoE rewards that explicitly penalize MTP latency spikes and frame drops; and handover-aware prediction using radio telemetry [2], [12]. The proposed methodology integrates LSTM forecasting with A3C to predict degradation and adjust bitrate proactively, prioritizing responsiveness during high-motion gameplay and cell transitions [2], [10].

Chapter 3 Methodology

3.1 System Design

The proposed system follows a trace-driven simulation approach, where real-world 5G network measurements are used to train and evaluate a Long Short-Term Memory (LSTM) neural network for proactive throughput prediction. Rather than relying on synthetic network simulation, the system leverages the LUMOS 5G dataset, which captures authentic mmWave signal behavior across multiple mobility scenarios in a live commercial 5G deployment [10]. The predicted throughput values are then passed to a bitrate adaptation module that selects the appropriate video quality level before network degradation reaches the client.

The system pipeline consists of four sequential stages. First, raw 5G telemetry data is loaded from the LUMOS dataset and preprocessed into a normalized feature matrix. Second, overlapping time-series sequences are constructed from this matrix to serve as training input for the LSTM model. Third, the trained LSTM predicts the next throughput value given the ten most recent time steps of network measurements. Finally, the predicted throughput is mapped to one of five video quality levels ranging from 360p to 1440p Ultra HD through a threshold-based bitrate adaptation policy.

This pipeline directly addresses the core limitation of reactive ABR systems identified in the literature that bitrate adjustment only occurs after throughput has already collapsed [1]. By predicting throughput one step ahead using temporal signal patterns, the system enables pre-emptive bitrate switching that protects Motion-to-Photon latency during volatile 5G conditions and cell tower handover events.

3.2 Hardware and Software Components

We built the system using Python inside a virtual environment on a regular personal computer that runs Windows 11. We did not need any GPU hardware because the size of the LUMOS dataset and the complexity of the model could be handled by a CPU alone. We chose PyTorch as the deep learning framework because it works well with Python 3.14 and its flexible dynamic computation graph makes it easier to create and debug LSTM architectures.

TABLE I
HARDWARE AND SOFTWARE SELECTION SUMMARY

Tool	Function	Alternatives	Selection Reason
Python 3.14	Primary programming language	R, MATLAB	Free with extensive ML library support
PyTorch 2.12	LSTM model construction and training	TensorFlow, Keras	Native Python support and flexible architecture
Pandas 3.0	Dataset loading and preprocessing	NumPy, Polars	Efficient CSV handling for large datasets
NumPy 2.5	Numerical computation and array operations	SciPy	Core dependency for ML operations
Matplotlib 3.11	Result visualization and graph generation	Seaborn, Plotly	Publication-quality static plots
Scikit-learn 1.9	Data normalization and evaluation metrics	StatsModels	MinMaxScaler and error metrics
Jupyter Notebook 7.6	Interactive development environment	VS Code, Spyder	Inline visualization during development
LUMOS 5G Dataset v1.0	Real-world 5G throughput traces	Irish 5G, MN-Wild	Public multi-mobility mmWave data

3.3 Hardware and Software Implementation

3.3.1 Dataset Description and Preprocessing

The LUMOS 5G dataset version 1.0 contains 68,118 measurement records collected from Verizon's mmWave 5G network in three locations in Minneapolis, Minnesota: a four-way traffic intersection, a mall area inside Minneapolis-St. Paul International Airport, and a 1,300-meter driving loop near U.S. Bank Stadium [10]. Each timestamped record describes the network conditions at that instant, including downlink throughput measured using iPerf 3.7, LTE and 5G New Radio signal strength, device speed, movement direction, and cell tower identity.

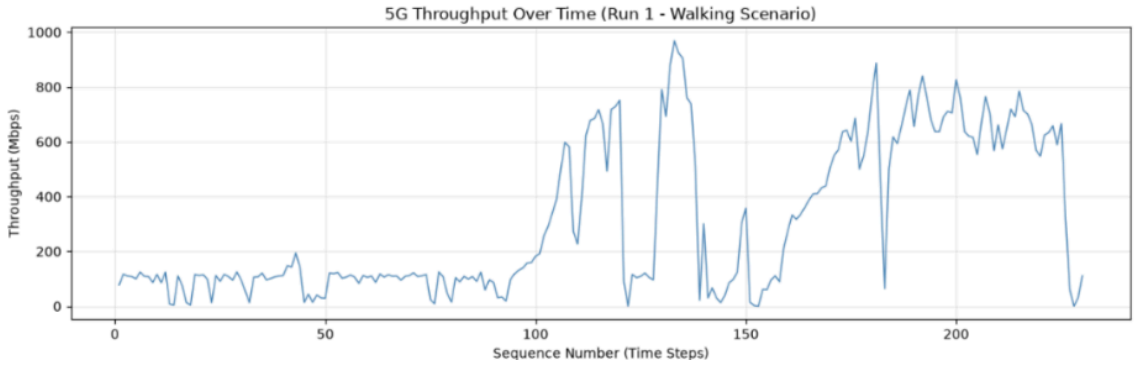


Fig. 4. Raw 5G downlink throughput measured over time for a single walking scenario trace from the LUMOS dataset, illustrating the volatile and unpredictable nature of commercial mmWave 5G signals.

Seven features directly related to throughput prediction were selected as model inputs: throughput, moving speed, LTE RSRP, LTE RSRQ, NR SS-RSRP, NR SS-RSRQ, and NR

SS-SINR. Missing values in the 5G-specific signal columns occurred when the device fell back to LTE; these values were replaced with zero to indicate the absence of a 5G NR connection. All seven features were then scaled to [0, 1] using Min-Max normalization to stabilize gradient computation during LSTM training.

A handover analysis was conducted by tracking changes in the tower_id field across consecutive records. A total of 1,371 handover events were identified, representing 2.01% of all time steps. Of these events, 881 occurred during walking scenarios and 490 occurred during driving scenarios. Average downlink throughput decreased by 85.23 Mbps immediately after a handover, from a pre-handover mean of 386.99 Mbps to a post-handover mean of 301.75 Mbps. This measured degradation supports the need for proactive prediction during cell transitions.

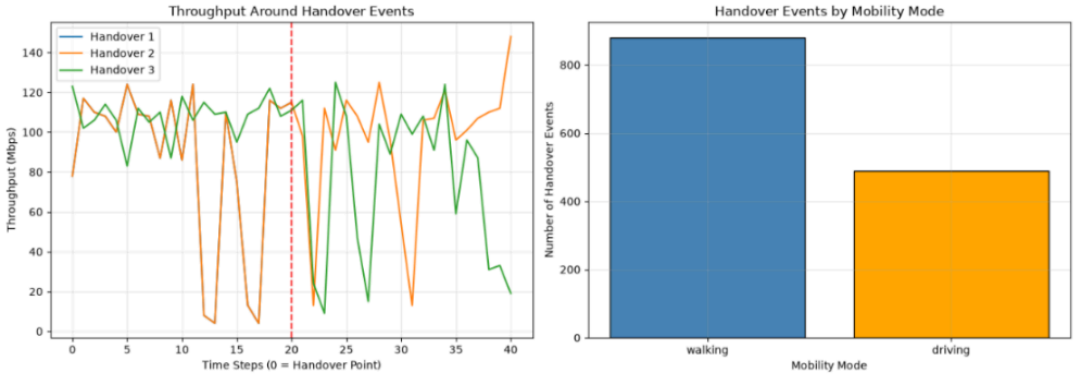


Fig. 5. Left: Downlink throughput fluctuation around three sample handover events (dashed line marks handover point). Right: Distribution of handover events across walking and driving mobility modes.

3.3.2 Sequence Construction and Data Splitting

A sliding-window approach was applied to the normalized feature matrix. Each input sample contains ten consecutive time steps across all seven features, and the target is the normalized throughput at the following time step. The ten-step window gives the LSTM enough recent context to identify emerging signal-fading patterns while keeping memory and computation requirements manageable. Applying this procedure produced 68,108 input-target sequence pairs.

The dataset was divided chronologically using an 80:20 ratio, resulting in 54,486 training samples and 13,622 testing samples. The samples were not shuffled, preserving the natural temporal order of the measurements and preventing future observations from leaking into the training set.

3.3.3 LSTM Model Architecture

The throughput predictor is a two-layer Long Short-Term Memory network implemented in PyTorch. LSTM input, forget, and output gates allow the model to preserve useful temporal

information while discarding irrelevant observations. This behavior is suitable for recognizing both abrupt signal changes and slower mobility-related fading patterns.

The first LSTM layer receives the seven input features over ten time steps and produces a hidden representation of size 64. A dropout rate of 0.2 is applied between the recurrent layers to reduce overfitting. The second LSTM layer refines the temporal representation, and its final hidden state is passed to a fully connected layer with 32 neurons and ReLU activation. A final single-neuron layer outputs the predicted normalized throughput for the next time step.

The model was trained with the Adam optimizer using a learning rate of 0.001 and Mean Squared Error (MSE) loss. MSE places greater emphasis on large prediction errors, which is useful because sudden throughput drops have the greatest effect on cloud-gaming quality. Training ran for 50 epochs with a batch size of 256, while PyTorch's DataLoader handled mini-batch generation efficiently.

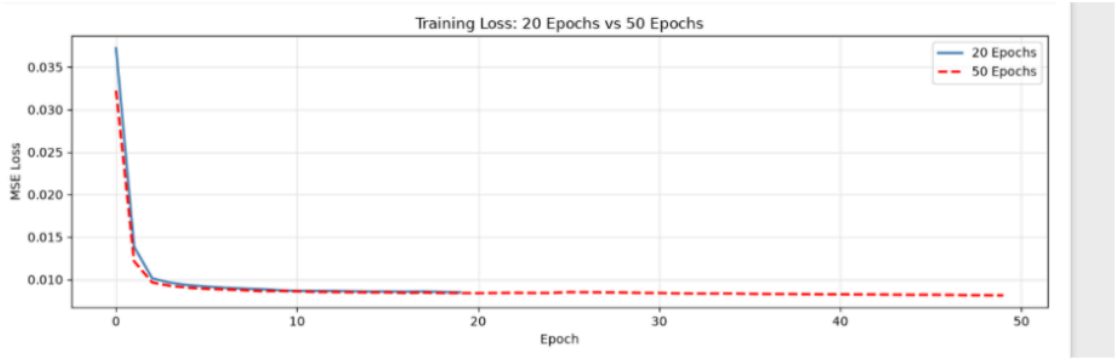


Fig. 6. Training loss over 20 and 50 epochs, demonstrating consistent convergence without overfitting across extended training.

3.3.4 Bitrate Adaptation Logic

The predicted normalized throughput is converted back to Mbps using the inverse Min-Max transformation and compared with five fixed thresholds. Predictions above 500 Mbps select 1440p Ultra HD at 6,000 Kbps, values from 200 to 500 Mbps select 1080p Full HD at 3,000 Kbps, values from 100 to 200 Mbps select 720p HD at 2,000 Kbps, values from 50 to 100 Mbps select 480p SD at 1,000 Kbps, and values below 50 Mbps select 360p Low at 500 Kbps.

The 1440p tier accounted for 40.5% of test predictions and occurred primarily in stationary or low-interference portions of the traces. Walking traces frequently produced predictions in the 200-500 Mbps range, while the 100-200 Mbps tier was common during periods with more frequent signal degradation. The two lowest tiers were triggered in fewer than 1% of predictions. Because the decision uses the LSTM forecast for the next time step rather than the currently observed

throughput, the system can reduce video quality before degradation reaches the client, helping prevent frame freezes and input lag caused by reactive adaptation.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

This chapter presents the experimental results obtained from training and evaluating the proposed LSTM-based proactive throughput prediction system on the LUMOS 5G dataset. The results are organized into five sections: overall model performance, comparison against baseline predictors, mobility-specific analysis, bitrate adaptation outcomes, and handover impact assessment.

4.1 Overall Model Performance

The LSTM model was trained on 54,486 sequences and evaluated on 13,622 test sequences drawn from the LUMOS 5G dataset. Table II summarizes the prediction error metrics for the 20-epoch and 50-epoch training configurations.

TABLE II
LSTM MODEL PERFORMANCE AT 20 AND 50 EPOCHS

Configuration	MSE	RMSE	MAE
LSTM 20 Epochs	0.007694	0.087716	0.056462
LSTM 50 Epochs	0.007447	0.086299	0.053223

Training for 50 epochs improved all three metrics compared with the 20-epoch model. Mean Squared Error (MSE) decreased from 0.007694 to 0.007447, a reduction of approximately 3.2%, while Mean Absolute Error (MAE) decreased from 0.056462 to 0.053223. The training-loss curves presented in Fig. 6 confirm that the model continued to converge after 20 epochs without showing evidence of overfitting, supporting the longer training duration. Unless otherwise stated, the subsequent analyses use the 50-epoch model.

Fig. 7 compares predicted and measured throughput over 300 consecutive test time steps. The model follows the overall signal trend closely and captures the principal peaks and drops with reasonable accuracy. Larger deviations occur during abrupt throughput collapses, reflecting the inherent unpredictability of mmWave links during short blockage or transition events [2], [12].

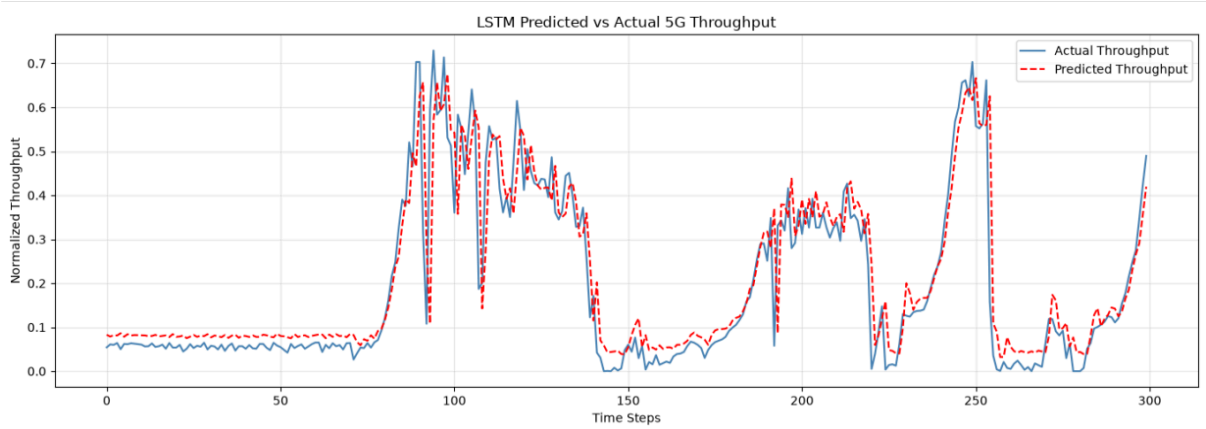


Fig. 7. LSTM-predicted throughput versus actual measured throughput over 300 test time steps, demonstrating the model's ability to track real 5G signal fluctuations.

4.2 Comparison Against Baseline Predictors

To evaluate the predictive advantage of the proposed model, it was compared with two standard baselines: a Simple Moving Average (SMA) predictor using a ten-step window and a Last Value predictor that repeats the most recently observed throughput. Table III presents the comparison.

TABLE III
COMPARISON OF LSTM AGAINST BASELINE PREDICTORS

Model	MSE	RMSE	MAE
LSTM (Ours)	0.007694	0.087716	0.056462
SMA Baseline	0.019840	0.140856	0.090253
Last Value	0.008210	0.090611	0.049969

The LSTM reduces MSE by 61.2% relative to the SMA baseline. SMA performs poorly because averaging the previous ten measurements introduces lag, preventing it from responding quickly to the sudden throughput peaks and drops common in mmWave environments. The Last Value predictor is competitive on MAE but performs worse on MSE and RMSE, indicating larger errors during abrupt signal changes where the LSTM's temporal memory provides an advantage.

Fig. 8 visually confirms the quantitative results. The SMA prediction lags behind rapid signal changes, whereas the LSTM prediction follows the actual throughput more closely across the test interval.

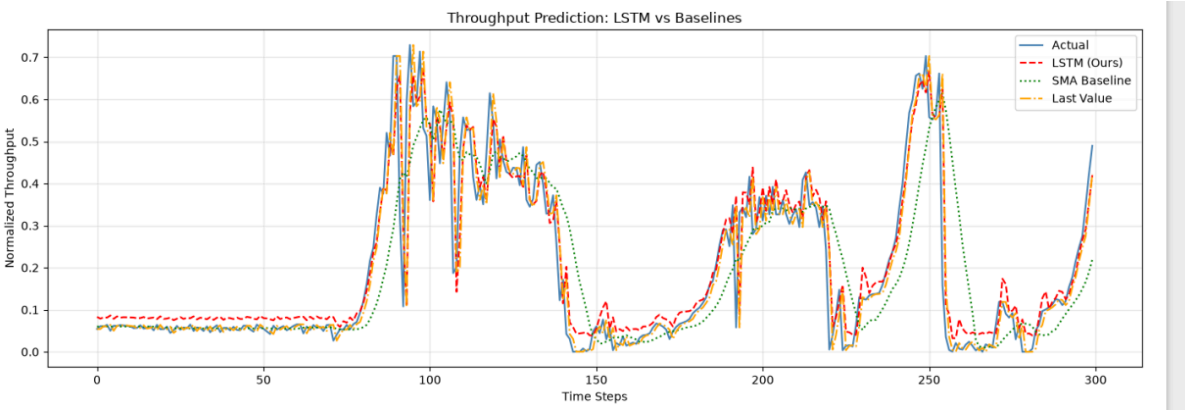


Fig. 8. Throughput prediction comparison between the LSTM, SMA baseline, and Last Value predictor over 300 test time steps against actual measured throughput.

4.3 Mobility-Specific Analysis

The test set was divided by mobility mode to determine whether prediction accuracy differs between walking and driving scenarios. Table IV summarizes the results.

TABLE IV
LSTM PREDICTION PERFORMANCE BY MOBILITY MODE

Mode	MSE	RMSE	MAE	Samples
Walking	0.008121	0.090116	0.058234	11,938
Driving	0.004668	0.068320	0.043902	1,684
Overall	0.007694	0.087716	0.056462	13,622

The model performs better in driving scenarios, where MSE is 42.5% lower than in walking scenarios. Driving traces generally follow consistent routes and exhibit smoother transitions between coverage zones, producing more predictable throughput sequences. Walking traces contain more directional changes and greater exposure to nearby buildings and obstacles, creating abrupt and irregular fluctuations that are difficult to predict from ten historical time steps alone.

This result has a practical implication for cloud gaming: pedestrians in dense urban environments encounter less predictable connectivity and therefore benefit more from proactive throughput prediction than users traveling along consistent vehicle routes.

4.4 Bitrate Adaptation Results

The bitrate adaptation module was applied to all test predictions. Table V summarizes the resulting distribution across the five quality tiers.

TABLE V
BITRATE TIER DISTRIBUTION ACROSS TEST PREDICTIONS

Quality Tier	Bitrate	Throughput Range	Samples	Percentage
1440p Ultra HD	6,000 Kbps	> 500 Mbps	5,491	40.3%
1080p Full HD	3,000 Kbps	200-500 Mbps	2,221	16.3%
720p HD	2,000 Kbps	100-200 Mbps	4,403	32.3%
480p SD	1,000 Kbps	50-100 Mbps	1,504	11.0%
360p Low	500 Kbps	< 50 Mbps	3	0.0%

The 1440p Ultra HD tier was selected for 40.3% of predictions, consistent with the high peak throughput available on Verizon's mmWave network in open areas. The 360p tier appeared only three times among 13,622 predictions, indicating that complete signal collapse was uncommon in the dataset. Overall, the system selected 720p or higher for 88.9% of predictions, maintaining a playable and visually acceptable quality level for most of the test session.

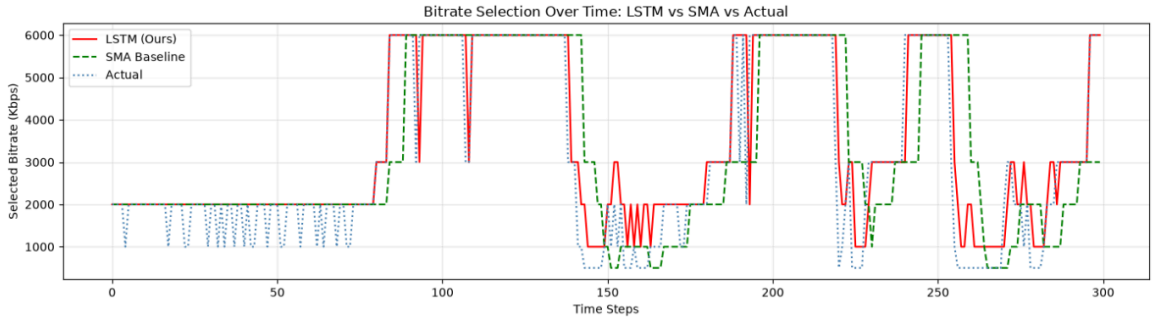


Fig. 9. Bitrate selection over 300 time steps comparing LSTM-driven adaptation with the SMA baseline and actual throughput-driven selection.

Table VI compares the frequency of bitrate changes produced by each method.

TABLE VI
BITRATE SWITCHING FREQUENCY COMPARISON

Method	Switching Frequency
LSTM (Ours)	19.34%
SMA Baseline	8.60%
Last Value	27.86%
Actual	27.87%

The LSTM produces a switching frequency of 19.34%, lower than both the Last Value predictor and the actual throughput-driven rate of 27.87%. The resulting policy avoids many unnecessary quality changes that could be distracting during gameplay. Although the SMA baseline switches least frequently, its stability results from prediction lag and substantially higher error; it can remain

in an unsuitable quality tier for an extended period. The LSTM provides a better balance between stability and responsiveness.

4.5 Handover Impact Analysis

A total of 1,371 handover events were identified in the complete dataset, representing 2.01% of all time steps. Walking scenarios accounted for 881 events and driving scenarios accounted for 490. Mean downlink throughput decreased by 85.23 Mbps immediately after a handover, from 386.99 Mbps before the transition to 301.75 Mbps afterward.

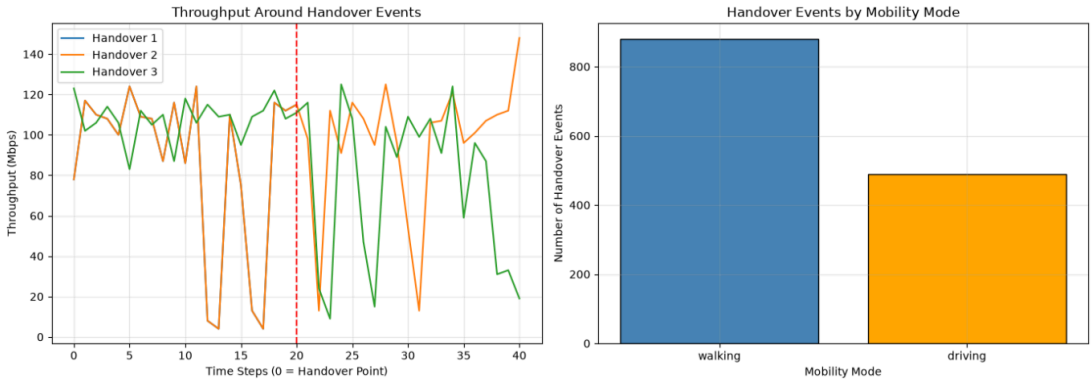


Fig. 10. Left: Throughput fluctuation around three sample handover events, with the red dashed line marking the transition point. Right: Handover event count by mobility mode.

If unaddressed, this 85.23 Mbps reduction would cause a reactive bitrate controller to lower quality only after degradation had reached the client and disrupted gameplay. By forecasting throughput one step ahead using recent throughput and radio-signal telemetry, the proposed LSTM enables the controller to begin downscaling before the reduction materializes. This result demonstrates the practical value of proactive prediction in handover-dense urban cloud-gaming scenarios.

Chapter 5 Conclusions

5.1 Summary

This research designed and evaluated an LSTM-based proactive throughput prediction system for adaptive bitrate streaming in cloud gaming over unstable 5G networks. Using the LUMOS 5G dataset, which contains 68,118 real-world mmWave measurements collected across walking and driving scenarios in Minneapolis, a two-layer LSTM network was trained to forecast downlink throughput one time step ahead from seven network-telemetry features.

After 50 training epochs, the model achieved an MSE of 0.007447, an RMSE of 0.086299, and an MAE of 0.053223. It reduced MSE by 61.2% relative to the Simple Moving Average baseline. The LSTM-driven adaptation policy also produced a lower bitrate-switching frequency than direct actual-throughput-driven adaptation, with rates of 19.34% and 27.87%, respectively. This result indicates that the proposed approach can provide more stable quality selection during volatile network conditions.

The handover investigation identified 1,371 cell-tower transitions and an average throughput reduction of 85.23 Mbps following each handover, highlighting the practical importance of predictive adaptation in urban cloud-gaming environments. Across all test predictions, the system maintained a quality level of 720p or higher in 88.9% of cases. These results demonstrate that proactive LSTM-based prediction can help sustain acceptable Quality of Experience (QoE) under the variable conditions of commercial mmWave 5G deployments.

5.2 Limitations

Several limitations should be acknowledged. First, the current model predicts throughput only one time step ahead. It therefore estimates the immediately following value rather than forecasting a degradation event 100-300 milliseconds in advance, as envisioned in the original research objective. Extending the prediction horizon would require a sequence-to-sequence or multi-step forecasting architecture, together with sufficient training data for each forecast interval.

Second, the bitrate adaptation module uses fixed throughput thresholds rather than a learned policy. Consequently, it cannot automatically adapt its decisions to different games, user preferences, or changing QoE priorities. A complete Deep Reinforcement Learning implementation based on A3C could replace the fixed mapping with a policy trained to maximize a cloud-gaming-specific reward.

Third, the LUMOS dataset does not include Motion-to-Photon latency or frame-drop measurements. The experiments therefore cannot directly quantify the effect of the proposed

method on end-user gaming QoE. Finally, the system was evaluated through trace-driven simulation rather than a live network deployment. This approach may not capture all operational effects, including packet loss, jitter, queuing delay, and inference overhead.

5.3 Future Improvement

The most immediate extension is to predict multiple future time steps, allowing the system to anticipate throughput degradation 200-300 milliseconds before it occurs. This forecast horizon would better align the predictor with the Motion-to-Photon latency constraints of interactive cloud gaming.

Replacing the threshold-based bitrate selector with an A3C agent that incorporates LSTM predictions into its state space would complete the proactive Deep Reinforcement Learning framework proposed in this research. The reward function could jointly consider visual quality, bitrate stability, frame loss, and interaction latency.

Future work should also incorporate game-content complexity as an input feature. For example, bitrate reductions during high-motion gameplay could receive a greater penalty than equivalent reductions during static scenes, making the QoE objective more context-aware. Evaluating the model on additional datasets, including the Irish 5G and MN-Wild datasets, would further test generalizability across network operators, radio environments, and geographic regions.

Finally, deploying the prediction module on a mobile-edge computing node near the 5G base station could reduce inference and communication latency. Such a deployment would provide a practical path toward real-time validation and eventual integration with operational cloud-gaming platforms.

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